MICHAEL LEHN, Ph.D.

Scientific Software Engineer · C/C++ · Compilers & LLVM · High-Performance Computing

Ulm, Germany · Open to relocation to the United States or Canada

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SUMMARY

Software engineer and mathematician (Ph.D.) with 20+ years of hands-on experience developing performance-critical C/C++ software. Expertise spans numerical algorithms and high-performance linear algebra, memory hierarchy and cache-aware algorithms, SIMD, multithreaded and distributed computing, CPU architecture, compiler construction, and low-level systems programming. Recent projects include cross-architecture BLAS micro-kernels (x86-64, ARM SME2, CUDA), event-driven FPGA control for quantum experiments, and an LLVM-based compiler.

TECHNICAL SKILLS

Languages: C++ (modern, templates/compile-time techniques), C, Python, Assembly (x86-64, AArch64, custom RISC), LLVM IR, SystemVerilog, Java, Perl

Performance & HPC: Numerical linear algebra, BLAS, LAPACK, ScaLAPACK, SIMD micro-kernels (SSE3, NEON, SME2), cache-aware/blocked algorithms, std::thread, OpenMP, MPI, CUDA/Tensor Cores, benchmarking and profiling

Compilers & Systems: LLVM, compiler construction, code generation, x86-64 System V ABI, C interoperability, ISA design, CPU architecture, FPGA (iCE40, Artix-7/Sinara), bare-metal programming

Tools: Linux/Unix, macOS, Git, CMake, GDB, Valgrind, perf, Intel VTune, NVIDIA Nsight, Verilator, cocotb, Yosys

SELECTED PROJECTS (OPEN SOURCE)

ulmBLAS **— High-Performance BLAS Library** C++ · Assembly · CUDA

- Designed a complete BLAS library (levels 1–3, four data types) on the layered BLIS architecture: portable C++ blocking and packing with replaceable micro-kernels; tested with the Netlib BLAS and Reference-LAPACK suites.
- Wrote SSE3, NEON, and SME2 kernels and integrated BLIS, AOCL, and OpenBLAS kernels; with OpenBLAS's kernel, single-threaded DGEMM reaches ~97% of OpenBLAS and ~99% of oneMKL (Ice Lake, n ≥ 1000).
- Experimental Apple M5 SME2 backend: DGEMM up to 490 GFLOP/s, 1.06–1.15× Apple Accelerate for n ≥ 1050; Accelerate is still faster for medium sizes.
- CUDA DGEMM tutorial (in progress): Tensor Core kernels reach 0.82 of cuBLAS on an NVIDIA A100 (n ≥ 1000).
- Authored the GEMM Tutorial (referenced by FLAME); ulmBLAS is used in Meta Quest software.

ABC **("A Better C") — Systems Language & LLVM Compiler** C++ · LLVM

- Designed a small systems language exposing C's machine model with simplified declaration syntax; built an LLVM-based compiler with interchangeable backends targeting LLVM IR and custom ULM assembly.
- Implemented native C interoperability with x86-64 System V ABI lowering, enabling direct use of existing C libraries without wrappers; demonstrated with raylib applications.
- Developed not-abc, a minimal self-hosting compiler originally implemented in ABC, demonstrating the language's suitability for systems programming and compiler construction.

FLENS **— Flexible Library for Efficient Numerical Solutions** C++

- Built a numerical computing framework combining MATLAB-like abstractions with direct access to optimized BLAS/LAPACK kernels, achieving zero abstraction overhead as verified by benchmarks and generated assembly.
- Designed a generic architecture allowing algorithms to operate unchanged on local, distributed, GPU-resident, and implicitly represented matrices.
- Used compile-time C++ techniques and expression analysis to map high-level operations directly to optimized numerical kernels and detect performance-critical temporaries.

ulm-on-iCE / ulm-generator **— FPGA CPU & ISA Toolchain** SystemVerilog · C/C++ · Assembly

- Designed and implemented a custom RISC CPU in SystemVerilog on an iCE40 FPGA, from digital logic and ISA design to assembly programs running on physical hardware.
- Defined a declarative ISA description format and built a generator producing an assembler, virtual machine, instruction-level debugger, and documentation from a single specification.

qubic-notes **— QubiC Compiler, ISA & RTL Simulation** Python · Verilog · cocotb

- Studied LBNL's open-source QubiC qubit-control system; documented its compiler and ISA and built QD, a prototype quantum-control DSL compiled through QubiC-IR and the QubiC toolchain to machine code.
- Explored explicit timing and synchronization semantics (parallel joins, channel placement, ordering after runtime loops); verified only in cycle-accurate RTL simulation of QubiC's cores, FPROC, and shot sequencer.

PROFESSIONAL EXPERIENCE

Ulm University — Institute of Numerical Mathematics · Ulm, Germany 2008 – present

Tenured Faculty Member (Akademischer Rat auf Lebenszeit), Numerical Mathematics

- Develop and maintain the open-source projects above and own their full engineering lifecycle: architecture, implementation, testing, performance benchmarking and profiling, documentation, and long-term maintenance.
- Design and teach courses in high-performance computing, compiler construction, systems programming, numerical mathematics, and scientific computing; the HPC curriculum progresses from memory hierarchy and cache-aware algorithms through SIMD micro-kernels and shared-memory parallelism to MPI and GPU computing.
- Built a full-stack computing curriculum spanning logic gates, registers and ALUs, ISA design, assembly, systems programming, and compiler construction, supported by custom-developed software and FPGA hardware.

Ulm Center for Scientific Computing (UZWR), Ulm University · Ulm, Germany 2006 – 2008

Managing Director

- Developed complete C++/Qt applications for industrial and research partners.
- Translated industrial and interdisciplinary partners' engineering and scientific problems into mathematical models and working numerical software; developed custom solutions where no suitable off-the-shelf software existed.
- Among numerous interdisciplinary projects, designed and developed a self-contained Windows application for MRI-based biomechanical analysis, implementing the complete software stack in-house—from FEM and numerical algorithms to data processing and the graphical user interface.
- Scoped and evaluated incoming projects, designed computational approaches, and either implemented solutions directly or coordinated them with researchers possessing the required expertise.

Ulm University — Institute of Numerical Mathematics · Ulm, Germany 2002 – 2008

Research Associate

- Conducted research and software development in scientific computing and numerical linear algebra; developed FLENS as the core of my doctoral research on efficient, abstraction-preserving numerical software in C++.

University of Southern California — Modeling and Simulation Laboratory · Los Angeles, CA 1999 – 2001

Research Assistant / Teaching Assistant

- Conducted research in applied mathematics and scientific computing with Prof. Chunming Wang in USC's Modeling and Simulation Laboratory while completing an M.S. in Applied Mathematics; master's research conducted in collaboration with NASA's Jet Propulsion Laboratory (JPL).

SELECTED INDUSTRY & RESEARCH COLLABORATIONS

German Aerospace Center (DLR) — Real-Time Control / Quantum Technologies 2025 – present

- Collaborate with the DLR Institute of Quantum Technologies on real-time control of a space-ready atomic-gas quantum memory; analyzed the ARTIQ/Sinara architecture and proposed FPGA logic for latency-critical reactions.
- Develop HeraldRT, an event-driven SystemVerilog controller for Sinara Kasli (Artix-7): a herald photon switches lasers off via a processor-free hardware path; a synchronous path time-stamps and sequences the experiment.
- Simulated (800 self-checks), synthesized, placed, and routed with the open-source openXC7 flow, and generated a bitstream; hardware tests, latency and jitter measurements, and Vivado timing sign-off are pending.

Tensor AI Solutions — HPC / Tensor Networks 2024

- Collaborated with Tensor AI Solutions on the design and development of Tartarus, a C++ HPC library for efficient tensor contractions in tree-tensor-network applications for machine learning and QML.
- Co-supervised a research-driven student project, contributing substantially to the numerical algorithms, performance-oriented design, and development of the library.
- Applied TTGT and Loops-over-GEMM (LoG) contraction schemes, BLAS kernels (GEMM, GEMV, GER, DOT), memory-layout-aware algorithms, and Python-based C++ code generation to optimize CPU tensor contractions.

Daimler TSS (Daimler Group) — Machine Learning / Recurrent Neural Networks 2017 – 2018

- Collaborated with Daimler TSS on the mathematical analysis of training recurrent neural networks (vanishing and exploding gradients); co-supervised a research-driven bachelor's thesis.

EDUCATION

Ph.D., Numerical Mathematics — Ulm University, Germany 2008

Dissertation: FLENS — A Flexible Library for Efficient Numerical Solutions · Advisor: Prof. Karsten Urban

M.S., Mathematics — Ulm University, Germany 2002

M.S., Applied Mathematics — University of Southern California, Los Angeles, CA 2001